

Supplementary Material for: Expiry Moves the Close, Not the Day: Natural Experiments from India's Changing Options Expiry Calendar

This supplement reports the full expiry calendar and additional results referred to in the main text. All estimates use the specifications of the main text unless stated otherwise. Standard errors clustered by calendar week are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels.

S1. Expiry rules of Indian index options, 2014–2026

Table S1. Expiry rules by product

Product	From	Weekly	Monthly	Note
NIFTY	before 2014	-	last Thu	monthly contracts only
NIFTY	2019-02-11	Thu	last Thu	weekly options launched 11 Feb 2019 (Thursday)
NIFTY	2025-09-01	Tue	last Tue	NSE moves all expiries to Tuesday (last Thursday expiry 28 Aug 2025)
BANKNIFTY	before 2014	-	last Thu	monthly contracts only
BANKNIFTY	2016-05-27	Thu	last Thu	weekly options launched 27 May 2016 (Thursday)
BANKNIFTY	2023-09-04	Wed	last Thu	weekly moved to Wednesday (first 6 Sep 2023); monthly stays last Thursday
BANKNIFTY	2024-03-01	Wed	last Wed	monthly moved to last Wednesday (first 27 Mar 2024)
BANKNIFTY	2024-11-20	-	last Wed	weekly contracts discontinued (SEBI; last weekly 13 Nov 2024)
BANKNIFTY	2025-01-02	-	last Thu	monthly moved to last Thursday (NSE circular, 29 Nov 2024)
BANKNIFTY	2025-09-01	-	last Tue	monthly moved to last Tuesday
SENSEX	2023-05-15	Fri	last Fri	relaunched 15 May 2023 with Friday expiry
SENSEX	2025-01-04	Tue	last Tue	moved to Tuesday from 1 Jan 2025 (last Friday expiry 3 Jan, first Tuesday 7 Jan)
SENSEX	2025-09-01	Thu	last Thu	moved to Thursday from 1 Sep 2025
FINNIFTY	2021-01-11	Thu	last Thu	launched 11 Jan 2021 (Thursday)
FINNIFTY	2021-10-15	Tue	last Tue	moved to Tuesday (first 19 Oct 2021)
FINNIFTY	2024-11-20	-	last Tue	weekly contracts discontinued (last weekly 19 Nov 2024)
FINNIFTY	2025-01-02	-	last Thu	monthly moved to last Thursday
FINNIFTY	2025-09-01	-	last Tue	monthly moved to last Tuesday
MIDCPNIFTY	2022-01-24	Tue	last Tue	launched 24 Jan 2022 (Tuesday)
MIDCPNIFTY	2023-08-17	Mon	last Mon	moved to Monday (first 21 Aug 2023)
MIDCPNIFTY	2024-11-19	-	last Mon	weekly contracts discontinued (last weekly 18 Nov 2024)

Product	From	Weekly	Monthly	Note
MIDCPNIFTY	2025-01-02	-	last Thu	monthly moved to last Thursday
MIDCPNIFTY	2025-09-01	-	last Tue	monthly moved to last Tuesday
BANKEK	2023-05-15	Fri	last Fri	relaunched 15 May 2023 with Friday expiry
BANKEK	2023-10-16	Mon	last Mon	moved to Monday from 16 Oct 2023
BANKEK	2024-11-19	-	last Mon	weekly contracts discontinued (last weekly 18 Nov 2024)
BANKEK	2025-01-01	-	last Tue	monthly moved to last Tuesday
BANKEK	2025-09-01	-	last Thu	monthly moved to last Thursday

Note. NIFTY, BANKNIFTY, FINNIFTY (Nifty Financial Services) and MIDCPNIFTY (Nifty Midcap Select) trade on NSE; SENSEX and BANKEK on BSE. Each rule applies from its start date until the next rule of the same product. If the scheduled day is a holiday, contracts expire on the previous trading day. In the week of a monthly expiry, no separate weekly contract is listed.

S2. Alternative whole-session volatility measures

Table S2. Pooled estimates of equation (1) for alternative outcomes (daily data, 2014–2026)

	Garman-Klass	Parkinson	Rogers-Satchell	open-to-close return	close-to-close (incl. overnight)	overnight gap
Own weekly expiry	−0.024 (0.032)	−0.007 (0.036)	−0.042 (0.033)	0.161* (0.086)	−0.007 (0.032)	0.051 (0.076)
Own monthly expiry	0.055 (0.049)	0.048 (0.053)	0.058 (0.052)	0.103 (0.116)	−0.007 (0.050)	−0.299** (0.121)
Another large index expires	−0.062* (0.034)	−0.049 (0.038)	−0.074** (0.037)	0.015 (0.084)	−0.065* (0.037)	−0.033 (0.082)
Small index expires	−0.094* (0.048)	−0.120** (0.053)	−0.073 (0.050)	−0.251** (0.110)	−0.086* (0.048)	0.052 (0.106)

Note. Parkinson and Rogers–Satchell are range-based variance estimators (Parkinson, 1980; Rogers & Satchell, 1991); open-to-close is the squared session return; close-to-close adds the squared overnight gap; overnight gap is the squared return from the previous close to the open.

S3. Pinning

Table S3. Closing values near strikes, and moves toward the largest open-interest strike

Index	Strike step	Share near a strike: expiry days	Other days	Weekly expiry coefficient	Monthly expiry coefficient
Nifty 50	50	0.208	0.195	−0.003 (0.033)	−0.022 (0.042)
Nifty 50	100	0.211	0.184	0.028 (0.034)	−0.048 (0.041)
Bank Nifty	100	0.228	0.199	0.075** (0.032)	0.107** (0.044)
Bank Nifty	500	0.228	0.185	0.015 (0.033)	0.064 (0.047)
Sensex	100	0.244	0.201	0.042 (0.047)	0.038 (0.080)
Sensex	500	0.202	0.187	−0.005 (0.043)	−0.111 (0.071)

Index	Expiry days	Mean move, expiry days (%)	Mean move, other days (%)	Difference, controlling for distance at open
Bank Nifty	486	0.222	0.152	0.098 (0.066)
Nifty 50	443	0.201	0.110	0.107** (0.052)

Note. Top panel: share of official closing values whose distance to the nearest strike (a multiple of the strike step) is less than 10% of the step. Without pinning this share is about 0.20. The coefficients are from linear-probability versions of equation (1) for each index. Bottom panel (Nifty 50 and Bank Nifty, NSE contract files 2014 – June 2026): change during the day in the distance, in percent of the index level, between the index and the strike with the largest open interest (calls plus puts) at the previous close in the nearest-expiry series. Positive values mean the index moved away from that strike. The difference controls for the distance at the open, with weekday and year fixed effects.

S4. Intraday reversals

Table S4. Reversal of earlier moves on expiry days (one-minute data)

Sample	Afternoon on morning × expiry	Last half hour on rest of day × expiry	Days
All	0.090 (0.078)	0.007 (0.019)	3,428
Nifty 50	0.091 (0.112)	−0.004 (0.029)	1,248
Bank Nifty	0.072 (0.102)	0.031 (0.030)	1,252
Sensex	0.158 (0.109)	−0.025 (0.029)	928
Bank Nifty, Jan 2023 – Mar 2025	−0.001 (0.128)	−0.094*** (0.035)	551

Note. First column: coefficient on the interaction of the morning return (09:15–12:30) with own expiry in a regression of the afternoon return (12:30–15:30). Second column: coefficient on the interaction of the return from the open to 15:00 with own expiry in a regression of the return from 15:00 to 15:30. Negative values indicate stronger reversal on expiry days. The final row covers the period of the allegations in SEBI’s interim order of 3 July 2025.

S5. Settlement-window effects by period

Table S5. Expiry effects on 15:00–15:30 log realized variance by period

Period	Own weekly	Own monthly	Another large index
2019-23 (minute data: 2021-23)	0.192*** (0.056)	0.383*** (0.093)	−0.000 (0.075)
2024 to SEBI measures	0.603*** (0.186)	0.714*** (0.180)	0.226 (0.190)
SEBI measures to interim order	0.494** (0.187)	0.312** (0.131)	0.258 (0.152)
after interim order	0.921*** (0.169)	0.396*** (0.106)	0.489*** (0.162)

Period	Nifty 50	Bank Nifty	Sensex
2019-23 (minute data: 2021-23)	0.011 [136] (0.051)	0.151** [136] (0.066)	0.512*** [33] (0.131)
2024 to SEBI measures	0.360* [46] (0.192)	0.521*** [46] (0.161)	1.037*** [46] (0.182)
SEBI measures to interim order	0.230 [32] (0.195)	0.615** [8] (0.263)	0.566*** [33] (0.152)

Period	Nifty 50	Bank Nifty	Sensex
after interim order	0.552*** [51] (0.152)	0.188 [12] (0.220)	1.121*** [51] (0.157)

Note. Top panel: pooled estimates. Bottom panel: own-expiry effect for each index (number of own expiry days in brackets). Both compare indices with each other on the same weekday and week (index-by-week and common weekday fixed effects). SEBI's measures took effect on 20 November 2024; the interim order is dated 3 July 2025. One-minute data start in May 2021, so the first period covers 2021–2023.

S6. Forecasting next-day volatility

Table S6. Does the expiry calendar improve volatility forecasts?

Index	Forecast days	QLIKE, HAR	QLIKE, HAR + calendar	Change (%)	Diebold–Mariano <i>p</i>
Nifty 50	2,107	0.3702	0.3754	+1.4	0.039
Bank Nifty	2,109	0.3581	0.3645	+1.8	0.062
Sensex	2,105	0.3756	0.3792	+1.0	0.013
Midcap 50	2,100	0.3468	0.3474	+0.2	0.803

Note. Walk-forward forecasts of next-day close-to-close variance from January 2018 to July 2026, with an expanding estimation window re-fitted every 22 trading days. The HAR model (Corsi, 2009) uses log variance over the last day, week and month and the calendar days to the next session, and is estimated with the Gamma loss, which is consistent with the QLIKE criterion (Patton, 2011). Diebold–Mariano tests (Diebold & Mariano, 1995) use Newey–West standard errors. “+ calendar” adds the next day’s own weekly, own monthly and other-large-index expiry indicators, which are known in advance. Lower QLIKE is better.

S7. The closing auction: a first look

Table S7. Expiry effects in the hourly bar starting at 15:15 (the last bar of the session), before and after the closing auction

	Own expiry	Another large index expires	Own expiry × after 3 Aug 2026	Expiry days after
All	0.38*** (0.08)	0.21*** (0.07)	0.99** (0.41)	21
Nifty 50	0.44*** (0.10)	0.29*** (0.07)	0.16 (0.74)	10

	Own expiry	Another large index expires	Own expiry × after 3 Aug 2026	Expiry days after
Bank Nifty	0.61*** (0.10)	0.13 (0.10)	0.35 (1.30)	2
Sensex	0.22 (0.16)	0.17 (0.14)	1.97*** (0.74)	9

Note. Log Parkinson variance of the hourly bar starting at 15:15, October 2023 – September 2026 (Yahoo Finance). “After 3 Aug 2026” covers about two months. The hourly bar may not capture the auction price itself, and few expiry days fall after the change, so the test has little power.

S8. Total weekly volatility

Table S8. Weekly options and total weekly variance

Measure	Weekly-options regime	<i>p</i>	Index-weeks
close-to-close	0.054 (0.017)	0.002	2,510
session (Garman-Klass)	0.056 (0.017)	0.001	2,510
close-to-close, + index-specific trends	−0.018 (0.017)	0.299	2,510

Note. Index-week panel. The outcome is the log of average daily variance in the week; the regressor indicates that the index’s own options had weekly contracts. Index and week fixed effects; the last row adds index-specific linear trends.

S9. Expiry-day option trading

Table S9. Share of each year’s option contracts traded on expiry day in the expiring series (%)

Year	Nifty 50	Bank Nifty
2014	9.1	13.7
2015	9.3	12.7
2016	10.5	34.3
2017	11.1	41.5
2018	10.0	43.7
2019	25.4	44.5
2020	32.8	45.9
2021	32.0	41.4
2022	36.0	43.9

Year	Nifty 50	Bank Nifty
2023	45.3	50.0
2024	59.0	62.6
2025	46.5	38.5

Source. NSE F&O contract files.

S10. Additional robustness checks

Table S10. Additional checks of the settlement-window result

Check	Outcome	Coefficient	<i>p</i>	Observations
stacked events, no control — weekday gaining an expiry	15:00–15:30, 1-min RV	0.503*** (0.087)	0.000	1,366
stacked events, no control — weekday losing an expiry	15:00–15:30, 1-min RV	−0.375*** (0.102)	0.000	1,366
stacked events, + other large index expires — weekday gaining an expiry	15:00–15:30, 1-min RV	0.560*** (0.099)	0.000	1,366
stacked events, + other large index expires — weekday losing an expiry	15:00–15:30, 1-min RV	−0.442*** (0.109)	0.000	1,366
stacked events, no control — weekday gaining an expiry	whole session, 5-min RV	0.143** (0.069)	0.040	1,366
stacked events, no control — weekday losing an expiry	whole session, 5-min RV	−0.038 (0.098)	0.698	1,366
stacked events, + other large index expires — weekday gaining an expiry	whole session, 5-min RV	0.160** (0.075)	0.035	1,366
stacked events, + other large index expires — weekday losing an expiry	whole session, 5-min RV	−0.058 (0.114)	0.613	1,366
leave out nifty50 — own weekly expiry	15:00–15:30, 1-min RV	0.568*** (0.048)	0.000	2,180
leave out nifty50 — own monthly expiry	15:00–15:30, 1-min RV	0.534*** (0.065)	0.000	2,180
leave out banknifty — own weekly expiry	15:00–15:30, 1-min RV	0.514*** (0.054)	0.000	2,176
leave out banknifty — own monthly expiry	15:00–15:30, 1-min RV	0.635*** (0.084)	0.000	2,176

Check	Outcome	Coefficient	<i>p</i>	Observations
leave out sensex — own weekly expiry	15:00–15:30, 1-min RV	0.185*** (0.050)	0.000	2,500
leave out sensex — own monthly expiry	15:00–15:30, 1-min RV	0.344*** (0.078)	0.000	2,500
settlement window from 5-minute returns — own weekly expiry	15:00–15:30, 5-min RV	0.193*** (0.058)	0.001	3,428
settlement window from 5-minute returns — own monthly expiry	15:00–15:30, 5-min RV	0.466*** (0.097)	0.000	3,428
settlement window from 5-minute returns — another large index expires	15:00–15:30, 5-min RV	0.121** (0.058)	0.038	3,428
dose-response: own expiry x expiring-series contracts (std.) — own expiry (at average volume)	15:00–15:30, 1-min RV	0.183*** (0.043)	0.000	2,494
dose-response: own expiry x expiring-series contracts (std.) — own expiry × expiring-series contracts (1 s.d.)	15:00–15:30, 1-min RV	0.209*** (0.045)	0.000	2,494

Note. Stacked events: equation (2) pooled over the six calendar changes with one-minute data, without and with a control for the expiry of other large indices. Leave-one-out: pooled equation (1) without the named index. Five-minute returns: settlement-window realized variance from the six five-minute returns between 15:00 and 15:30. Dose–response (Nifty 50 and Bank Nifty, to June 2026): own expiry interacted with the standardized log number of contracts traded in the expiring series on the expiry day.

S11. Share of session variance in the settlement window

Table S11. Mean share of session realized variance in 15:00–15:30 (%)

Index	Own expiry days	Other days
Nifty 50	10.3	8.5
Bank Nifty	11.7	8.6
Sensex	21.1	8.9

S12. Additional figures

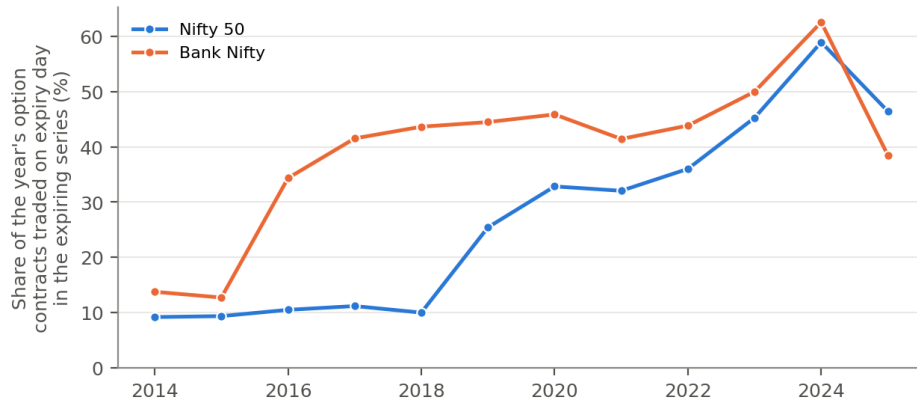


Figure S1. Share of each year's option contracts traded on expiry day in the expiring series, Nifty 50 and Bank Nifty options.

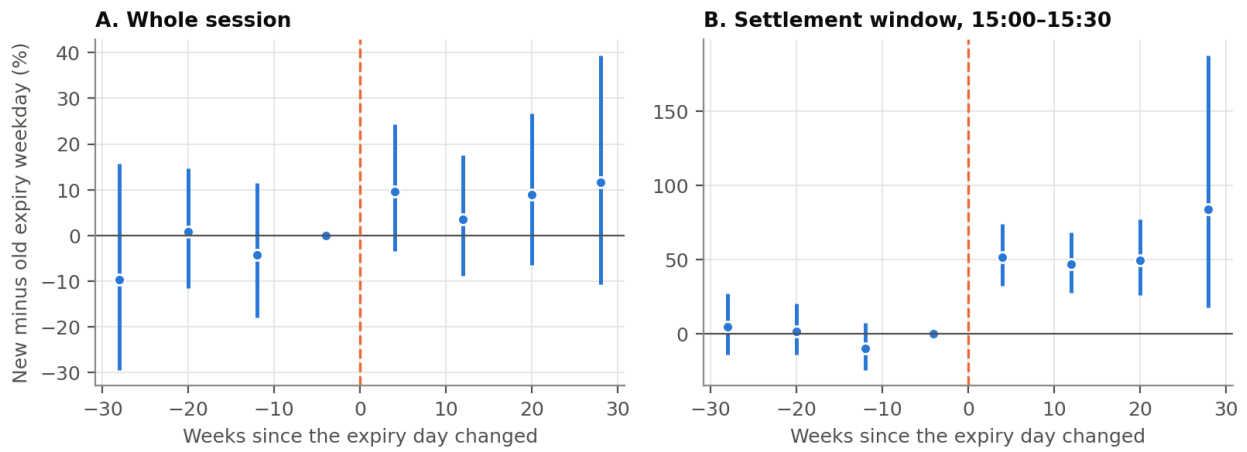


Figure S2. Event-time profile, stacked over the six calendar changes with one-minute data: difference between the new and the old expiry weekday in eight-week bins (the bin before the change is the reference), with 95% confidence intervals. A: whole session (five-minute realized variance). B: 15:00–15:30.



Figure S3. Whole-session volatility (log Garman–Klass variance) on each weekday relative to the rest of its week, 26 weeks before and after the eight major calendar changes. Unlike the settlement window (main text, Figure 1), the weekday profile does not consistently follow the expiry day.

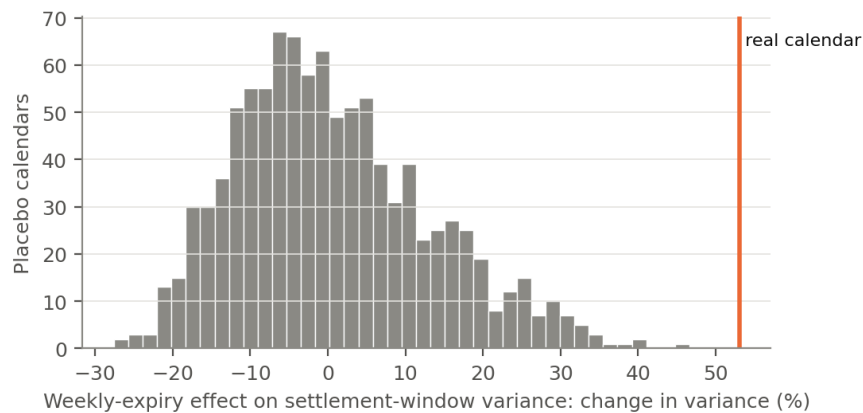


Figure S4. Distribution of the weekly-expiry effect on 15:00–15:30 variance under 1,000 placebo calendars; the vertical line is the estimate under the true calendar.

References

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